

Review

Pharmaceutical Wastewater and Sludge Valorization: A Review on Innovative Strategies for Energy Recovery and Waste Treatment

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Abstract: Currently, a large amount of pharmaceutical waste (PW) and its derivatives are being produced and, in some cases, inadequate management or treatment practices are applied. In this regard, this research explores the adoption of several alternatives to deal with these problems, including biocarbon within the framework of the circular economy. Photocatalytic nanomaterials have been also extensively discussed as a feasible way to remove pharmaceutical compounds in wastewater. Although there are existing reports in this area, this document provides a detailed study of the synthesis process, experimental conditions, the integration of photocatalysts, and their impact on enhancing photocatalytic efficiency. Additionally, the low cost and ease of fabrication of lab-scale microbial fuel cells (MFCs) are thoroughly examined. This innovative technology not only facilitates the degradation of hazardous compounds in wastewater but also harnesses their energy to generate electricity simultaneously. The aforementioned approaches are covered and discussed in detail by documenting interesting recently published research and case studies worldwide. Furthermore, this research is of significant importance because it addresses the valorization of PW by generating valuable by-products, such as H₂ and O₂, which can occur simultaneously during the photodegradation process, contributing to more sustainable industrial practices and clean energy technologies.

Keywords: pharmaceutical wastewater valorization; photocatalyst; electrical energy; fuel cell; water treatment; microbial fuel cell



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1. Introduction

Global pharmaceutical consumption is rising at an alarming rate, as a means of and concurrent with the need for improving life quality and managing seasonal illnesses and chronic diseases [1,2]. It has been reported that the annual rate of growth of the pharmaceutical market is 5.8% and is projected to exceed \$1.5 trillion by 2023 [3]. Recently, concern has been increasing regarding emerging Pharmaceutical Waste (PW) and the improper disposal of unused or expired drugs. These are commonly discharged into the environment, representing a serious threat to ecosystems and human health [4]. On the other hand, it must be highlighted that, although medical resources were limited during the COVID-19 pandemic, this situation accentuated the problem of PW, as both individuals and governments purchased large quantities of analgesics and antivirals to alleviate the symptoms of the virus [5]. PW is dumped with other solids residues to finally end up in landfills, dissolving and contaminating soil and water bodies. In addition, the high doses of medications make it almost impossible for the human body to fully absorb them. As a result, the active

substances or their metabolites are released into the environment via urine and feces [6]. This phenomenon was observed in the Baltic Sea where pharmaceuticals were found in groundwater [7]. The scientific community, in conjunction with governmental entities, have made several attempts to address the improper dumping of PW. Guidelines and policies for the industrial sector and even for citizens were proposed for the correct management of drug waste. However, it is important to note that, once this kind of waste is collected, an effective strategy needs to be adopted to recover and neutralize the active ingredients of PW. Unfortunately, conventional technologies are ineffective in removing those kinds of contaminants [8]. Incineration is the most widely used method for the treatment of PW because it is very easy to implement. Nonetheless, this strategy is too energy consuming considering that it requires temperatures above 1200 °C, and it is important to keep in mind that undesired harmful and toxic gases, such as CO₂, NO_x, and SO_x, are also released. Filtration and vertical biological reactors are mechanical systems commonly employed for water treatment; however, these technologies are not effective for pharmaceutical removal and can also be costly [9]. Meanwhile, conventional activated sludge is the most commonly used multi-chamber biological reactor for removing micropollutants. However, it has limited operating conditions due to temperature, pH, and low biomass concentration, which affect the growth of bacteria [10]. In this context, advanced processes have been proposed for degrading drugs including reverse osmosis, electrocatalysis, photocatalysis, plasma, and others [11]. Adsorption is a promising technique that has been used to reduce the concentration of micropollutants in wastewater and this can be more convenient if the concept of a circular economy is adopted [12]. Biochar has demonstrated good adsorption capacity for pharmaceutical waste and a lot of research has been done to understand this kinetic process [13]. For instance, Chauhan et al. conducted documental research based on biochar as an adsorbent material for the removal of pharmaceutical compounds, highlighting the importance of these materials for sustainable development [14].

At present, photocatalysts have emerged as an advanced and versatile approach for micropollutants treatment, and this has been successfully utilized to completely mineralize pharmaceuticals [15]. This alternative is advantageous considering that it uses low-cost materials and the activation energy can come from visible sunlight [16]. TiO₂, ZnO, and SnO₂ are semiconductors which are widely implemented to achieve advanced oxidation processes because they can use oxygen from the air, thus avoiding the use of complex oxidizing agents such as H₂O₂ or O₃ [17]. Recently, Ahmadpour et al. published a review of TiO₂-based photocatalysts as a versatile approach to removing pharmaceuticals from aquatic bodies [18]. They stated that a photocatalytic reaction involves several reactions. First, the target pollutant is transferred to the surroundings of the photocatalyst surface where adsorption takes place. Then, photocatalytic breakdown is achieved followed by the desorption of by-products. In the same direction, Muhammad et al. reported detailed research where silver-doped TiO₂ was studied as a photocatalyst for the demineralization of medicines and organic dyes [19]. Lin et al. developed a photocatalyst based on Ag₃PO₄ which was further modified with polypyrrole and multiwall carbon nanotubes, and then tested for tetracycline hydrochloride degradation, showing an efficiency of 100% [20].

One drawback that highlighted this group was the stability or recyclability of doped nanostructures, considering that the silver was unstable during the photocatalytic tests and because Ag was oxidized. These things considered, more experimental works are required to determine the optical load of silver in TiO₂ which ultimately creates lattice distortion.

On the other hand, MFCs have arisen as an innovative strategy aimed at the decomposition of organic molecules into CO₂, H₂O, and electrical energy as surplus, but there are still a lot of challenges to overcome before the widespread adoption of this technology [21]. It has been reported that microbial activity is influenced by the pH, temperature, and concentration of pollutants [22]. Thus, the scientific community is now dealing with the development of novel cell configurations and the progress of active anodes that assist the pharmaceutical degradation mechanism.

While some research has been conducted on pollutant removal, most studies have been limited to examining a single alternative or approach [23]. This research work discusses and analyzes in depth promising technologies for PW treatment, considering their efficiency, cost, and above all, versatility, when applied at a large scale. Photocatalysts seem to play a crucial role in the design of alternative devices capable of degrading PW. Therefore, they are considered in sections that explain in detail their chemical composition, synthesis methods, and physical and chemical features that need to be used as cost-effective material. Furthermore, the exploration of microbial fuel cells in degrading the active compounds in PW while at the same time producing electrical energy as a valued added subproduct, has also been considered. This study aims to provide a clearer pathway towards sustainable and economically feasible solutions for large-scale pollutant removal. The latter is crucial because this kind of technology deals with more than one sustainable development goal of the 2030 agenda, as proposed by the United Nations organization. Additionally, a conscientious analysis of regulations, policies, and guidelines regarding the management and disposal of PW is conducted. Finally, a future perspective and analysis of artificial intelligence trends and their impact on PW treatment are discussed.

2. Overview of Pharmaceutical Waste

Pharmaceutical contaminants are an important class of emerging pollutants, as these substances are potentially hazardous to biodiversity, even at low concentrations. In contrast to conventional pollutants, PWs exhibit resistance to degradation and can persist in ecosystems for extended periods, owing to the stability of their chemical structures. As a result, these substances can accumulate in water bodies, posing latent risks for both aquatic life and human health [24,25]. The presence of pharmaceutical compounds in water sources is mainly attributed to several factors, such as their widespread use for health care, veterinary uses, inadequate disposal practices, and ineffective water treatment processes [26]. Studies have indicated that conventional water treatment plants often have difficulty effectively removing pharmaceutical residues, resulting in contamination of freshwater reservoirs, rivers, and groundwater reserves [27]. Given the critical importance of clean water to sustain life and maintain ecosystems, it is essential to develop advanced treatment techniques capable of mitigating pharmaceutical contaminants. This can be observed in Figure 1. These treatment methods should aim to completely remove these pollutants to ensure the safety and integrity of water bodies. Different approaches have been investigated for the mitigation of pharmaceutical contaminants, including advanced oxidation processes, membrane filtration, activated carbon adsorption, and biological degradation. Each method offers unique advantages and challenges, highlighting the need for extensive research and innovation in the field of water treatment technology [28,29]. Furthermore, preserving water quality from pharmaceutical contaminants requires efforts from scientific communities, policymakers, and water management authorities. Thus, investing in research, enforcing regulations, and improving water treatment infrastructures is required if we want to obtain sustainable water systems that meet the needs of present and future generations [30–32].

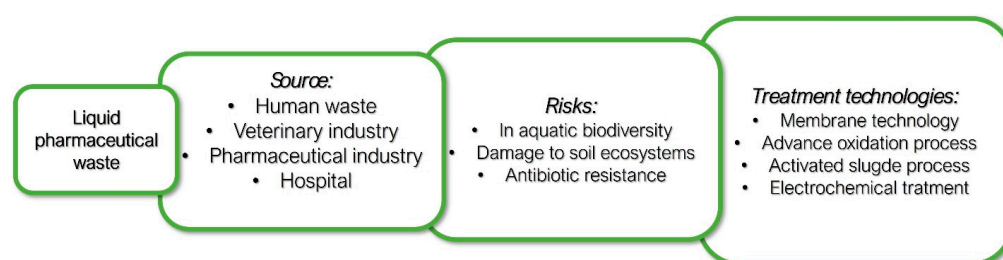


Figure 1. Overview of liquid pharmaceutical waste.

Sewage treatment normally yields sludge which is repurposed as fertilizer in agriculture. However, this practice poses a potential risk of pharmaceutical contamination,

which can be adsorbed by crops and end up in the food chain. Moreover, contaminants may leach into drinking water sources, demonstrating the critical need to address these environmental and public concerns [33]. The presence of antibiotics and antimicrobials in the environment also contributes to the development of drug-resistant pathogens, complicates the management of infections and diseases, and prompts the urgent need for treatment strategies capable of fully degrading such environmental pollutants [34]. Figure 2 illustrates the pathways through which water contaminants traverse various ecosystems, ultimately reaching humans.

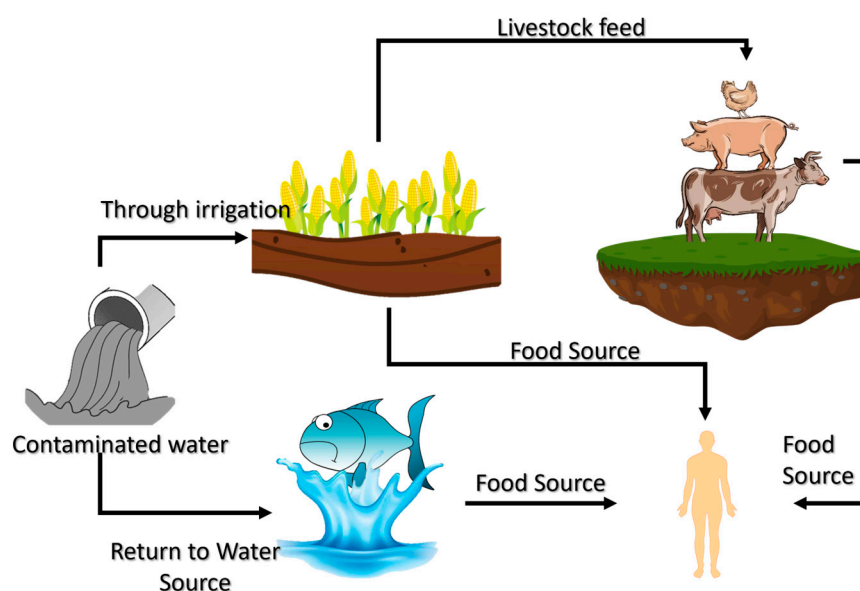


Figure 2. The pathways by which water contaminants move through diverse ecosystems, ultimately impacting human health.

Based on available data, water treatment plants have detected up to 43 different pharmaceutical compounds between 2010 and 2020 [35]. These compounds cover several categories, including antibiotics, anticonvulsants, antidepressants, estrogens, antidiabetics, lipid regulators, and various other pharmaceuticals. Among these, the pharmaceuticals with the highest concentrations found in daily discharges are acetaminophen, amoxicillin, and sulfamethoxazole [36]. Therefore, it is necessary to implement efficient treatments aimed at the complete degradation of these contaminants. Proper management of these compounds is crucial to protect the quality of drinking water and preserve long-term public and environmental health. It has been observed that, during the degradation of pharmaceuticals, their components release intermediate by-products that tend to be more toxic than the original molecule. Consequently, achieving complete decomposition of the original molecule and its by-products to mineralization, and converting them into CO_2 , H_2O , and inorganic ions is unpredictable [37]. With this purpose in mind, it is crucial to develop water treatment methods that are environmentally friendly and capable of effectively degrading these pollutants. In this context, advanced oxidation processes (AOPs) have proven to be a promising alternative to address this challenge.

AOPs are presented as complementary strategies that could enhance the efficiency of water treatment plants in the removal of pharmaceuticals. These processes involve the generation and utilization of highly reactive transient species, such as the hydroxyl radical ($\text{OH}^\cdot$). Due to its high oxidation potential, $\text{OH}^\cdot$ is considered one of the most potent oxidizing agents, capable of reacting with a wide range of organic contaminants. However, other oxygen species, such as the superoxide ion radical ($\text{O}_2^\cdot$) and the hydroperoxide radical ($\text{HO}_2^\cdot$), can also be used, although they have the disadvantage of being less active than $\text{OH}^\cdot$ [29,38]. There are several types of AOPs, such as ozone processes, chemical processes, electricity-based processes, and photo-assisted processes (see Table 1). The

main distinction between them lies in the route of formation of the OH^- radical. This radical is generally generated from water in the presence of a catalyst, as well as an energy source and/or chemical reagents [39]. Photocatalysis is one of the photo-assisted processes. Photocatalysis refers to the process by which a chemical reaction is accelerated by the activation of a material called a photocatalyst by ultraviolet, visible, or infrared radiation [40].

Table 1. Classification of AOPs.

Advanced Oxidation Processes (AOPs)			
Ozone	Chemical	Electricity-Based	Photo Assisted
Ozone in an alkaline medium	Fenton oxidation process	Electrochemical oxidation	Photo-assisted Fenton Process
Ozone with H_2O_2	Persulfate with heat	Electrocoagulation	Persulfate with UV
Ozone with UV and H_2O_2	Persulfate with Fe (II)	Electro-Fenton	Ozone with UV
Catalytic ozonation	Oxidation of persulfate		Peroxide with UV, Photocatalysis, Photoelectrocatalysis

2.1. Types of Pharmaceutical Waste

Liquid pharmaceutical contaminants are a growing concern due to their potential impact on human health and the environment. These contaminants can come from a variety of sources, including medical facilities, pharmacies, laboratories, and even private residences. The most common types of liquid pharmaceutical contaminants are expired or unused medications. It is estimated that between 10 and 25 tons of waste consisting of expired or unused medicines are generated per year for every million inhabitants [41]. These can include a wide range of pharmaceuticals, from intravenous solutions to syrups and disinfectant solutions, which may be improperly disposed, ending up in the water supply [41,42]. Antivirals account for 78.3% of the top disposed drugs, followed by hormones at 16.7% and diabetes medications at 11.5%. These values represent a significant increase compared to the year before the COVID-19 pandemic [43].

Kamal et al. [44] presented a study examining the disposal practices of expired and unused drugs, with participation from the USA, Italy, and Japan. Their research, based on interviews, yielded insightful data regarding the main reason for unused drugs, where the majority responded that their use of the medication was suspended because they no longer suffered from the disease. Conversely, a percentage mentioned that the medication was enough, and treatment was discontinued. The disposal method in the USA was to dispose of the medication in the toilet, according to a percentage of respondents (35.4%); in Italy, the most recurrent method was to return it to the pharmacy (51.2%), and in Japan it was to throw it in the trash (82.7%). This data confirms the need to establish standardized protocols for the proper disposal of medications.

Another liquid contaminant is laboratory and production waste. Research laboratories and pharmaceutical production facilities can generate large quantities of liquid waste during the pharmaceutical compound development, testing, and manufacturing processes. These wastes may contain synthetic chemicals in nanograms or micrograms per liter, organic solvents, and other compounds that can be hazardous if released into the environment without proper treatment. The pharmaceutical industry is the fastest-growing emerging sector globally, and it is estimated that this industry produces up to 3% of solid waste, while liquid effluent waste can contain up to 1000 micrograms per liter of active ingredients [45].

For the reduction of pollutants, industry actors apply prevention and reduction, in addition to considering government regulations. One significant approach involves substituting synthesis materials to minimize waste generation. Additionally, improvements to existing processes are continually implemented, with a focus on optimizing reaction

parameters to achieve maximum efficiency and minimize pollutant output. Controlling the conditions of reaction parameters leads to a reduction of pollutants obtaining a reaction efficiency of 100% and the implementation of the principles of green chemistry focused on the prevention of waste pollutants to the environment [43].

On the other hand, contaminated solutions or liquid pharmaceutical solutions can become polluted during the manufacturing, storage, or management processes. This may occur due to production errors, cross-contamination, or exposure to adverse environmental conditions. Contaminated solutions may contain active ingredients, chemical additives, or other components that may pose health risks if released into the environment. Contrast agents used in medical procedures, such as computed tomography and magnetic resonance imaging, can be significant sources of liquid pharmaceutical contamination [46]. These chemicals may contain heavy metals or other toxic substances that, if improperly disposed of, can contaminate bodies of water, and affect aquatic life [47]. Seitz et al. conducted a study to monitor iodinated contrast agents used in radiography in the Danube River in Germany [48]. The results of their investigation revealed peak concentrations of iodinated contrast agents in samples collected downstream of the metropolitan area. These concentrations, probably originating from local hospitals and radiological procedures, were detectable in surface water even after passing through wastewater treatment plants. The researchers concluded that iodinated contrasts are present in the natural waters of the Danube River. However, they emphasized that their assessment was limited in representativeness. It is important to note that, although significant concentrations of iodinated contrasts were identified, the study could not cover all potential sources of contamination or comprehensively assess their impact on the aquatic ecosystem. However, to mitigate the potential toxic effect of iodinated contrast compounds, new nanomaterials based on less polluting and more easily degradable metals, as well as assembled nanostructures, are being investigated. These advances in nanotechnology can offer safer and more sustainable alternatives for the application of contrast agents in X-rays and other medical procedures. Engineered nanomaterials with specific properties can help reduce the contaminant load in the environment while maintaining their clinical efficacy. Moreover, the enhanced degradability of some of these nanomaterials can help minimize their persistence in the environment and reduce potential adverse impacts on human health and the ecosystem [49].

Due to the insufficiency of normativity and a lack of knowledge, citizens throw unused, expired, or unwanted pharmaceutical items into municipal solid waste. Even in developed countries most of the collected trash ends up in municipal landfills where these items are leached. An interesting study was conducted by Zhang et al. who measured drug concentration in residual waste leachate and food waste in Shanghai [7]. They found that food waste has 14.5 mg d^{-1} of targeted pharmaceutical and personal care products where quinolone antibiotic was the main active drug. A deeper study can be found in [50] where the authors demonstrated the presence of 18 pharmaceuticals in the runoff samples in a residential area in Shanghai which was favored by the rainfall in the aquatic environment. To this end, time-resolved sampling of leaches in one municipal solid waste (MSW) sample was achieved during rainfall periods. Caffeine and Danofloxacin showed the highest concentration, representing a risk for aquatic organisms. The concern regarding leachate of PW has attracted global attention because this can be generated and persist even when the landfill is closed. A systematic study of leach samples, collected from 2019 to 2020, was conducted in four domestic landfills in North Carolina, United States [51]. Samples were tested to detect thirteen possible PW but only seven kinds of their degradants were positive. Reports indicated that anticonvulsants along with anti-inflammatories (carbamazepine and ibuprofen, respectively) were 100% detected. Antibiotics (amoxilin, sulfamethoxazole, etc.) are generally found at concentrations of 1000 ng/L in water sources, while analgesics (ibuprofen, paracetamol) are found below 1000 ng/L [52,53]. In this regard, waste-derived carbon dots are a promising alternative to face emerging pharmaceutical pollutants [54]. This type of material proves to have features capable of detection and degradation of PW.

2.2. Impact on Environment and Human Health

One of the main problems associated with pharmaceutical contaminants is their ability to bioaccumulate in living organisms. When these chemicals come in contact with the environment, they can be absorbed by aquatic organisms such as fish, crustaceans, and aquatic plants. As these contaminants move through the food chain, they can concentrate live organisms at higher levels and can be consumed by humans as part of their diet [55]. This can result in chronic exposure to low levels of pharmaceutical contaminants, increasing the risk of adverse health effects such as hormonal disorders [56], antibiotic resistance [57,58], and developmental problems [59]. Bexfield et al. [52] conducted the first large-scale systematic assessment of the presence of hormones and pharmaceuticals in groundwater intended for human consumption in the United States. In their study, a total of 21 hormones and 103 pharmaceuticals were identified. The results revealed that groundwater used as a source of drinking water in the U.S. is susceptible to contamination by these pharmaceutical compounds. Although the researchers noted that the concentrations detected did not appear to have adverse effects on human health, they did not address the bioaccumulation potential of these compounds, leaving this aspect for future research. Moreover, Fernando et al. conducted a study in one of five drinking water reservoirs in Canada, prompted by previous reports of elevated coliform levels [53]. In addition to confirming the presence of total coliforms, the study revealed the detection of several antibiotic resistance genes. The researchers emphasize the need for increased attention to proper water treatment on reserves, as the health of Canada's First Nations communities could potentially be at risk. Antibiotic concentrations have been reported globally, including: in Asia-Pacific countries: $450 \mu\text{g L}^{-1}$; in Africa: $50 \mu\text{g L}^{-1}$; in the Americas: $15 \mu\text{g L}^{-1}$; and in Europe: $10 \mu\text{g L}^{-1}$ [60].

Pharmaceutical contaminants not only affect human health but also have serious consequences for aquatic ecosystems. They can alter the endocrine systems of aquatic organisms and cause reproductive and developmental problems. In addition, these pollutants can alter the composition and diversity of biological communities by favoring pollutant-resistant species [61]. Moreover, pharmaceutical contaminants can affect terrestrial systems by leaching into the soil and contaminating groundwater [62]. To address these challenges, proactive measures are required to minimize the release of pharmaceutical contaminants into the environment and reduce their impact on human health and ecosystems. This may include implementing safe pharmaceutical waste management programs, promoting responsible prescribing and dispensing practices, and developing advanced treatment technologies to remove contaminants from wastewater and water bodies [63]. However, different methods have been used to obtain indicators of bioaccumulative toxicity, including ecotoxicity studies, risk analysis, and simulation models. All of this is meant to determine the negative impact of exposure to toxic pharmaceutical compounds on the environment, water sources, and human health.

2.3. Conventional Treatment Technologies

The effective removal of liquid pharmaceutical contaminants from the environment requires the development and application of advanced treatments and technologies. These solutions are designed to reduce or eliminate the presence of pharmaceuticals in water, thus mitigating negative effects on human health and ecosystems [64,65]. Conventional wastewater treatment systems may not be effective in completely removing pharmaceutical contaminants from water. Membrane filtration is a technique that includes ultrafiltration, nanofiltration, and reverse osmosis, and can be effective in removing pharmaceutical contaminants from water by trapping them in porous membranes [66]. One of the challenges hindering the advancement of nanofiltration is the low efficiency of existing membranes. Banjerdteerakul et al. conducted a study aimed to overcome this limitation in which they developed an innovative membrane composed of a thin film of covalent organic nanostructures (COF) applied on a predefined ceramic hollow fiber [67]. The results showed that this improved membrane exhibited high rejection against five environmentally persistent pharmaceuticals: diclofenac, sulfamethoxazole, ketoprofen, naproxen, and ibuprofen. In

addition to their outstanding filtration performance, this new generation of membranes is promising as a viable solution to address pharmaceutical compound contamination in wastewater. On the other side, activated carbon is widely used in water treatment systems to absorb organic contaminants, including pharmaceutical contaminants. Activated carbon has a large porous surface area that can trap and hold pharmaceutical compounds, effectively removing them from water [68]. Al-Sareji et al. carried out the development of activated carbon from pomegranate peel biomass, obtaining a high-quality, environmentally friendly, and cost-effective material [69]. This activated carbon was used to immobilize the laccase enzyme to remove pharmaceutical compounds such as diclofenac, amoxicillin, carbamazepine, and ciprofloxacin from wastewater. The study showed that the removal of these contaminants in aqueous media, with an initial concentration of 50 mg L^{-1} , was completed within 2 h. In addition, reuse tests were performed for more than six cycles, which demonstrated the stability and efficiency of the developed activated carbon. These results suggest that activated carbon derived from pomegranate peels could be an effective and sustainable solution for the removal of pharmaceutical compounds from wastewater. The main drawback of activated carbon is its limited stability and cyclability for pollutant absorption.

Some pharmaceutical contaminants can be degraded by microorganisms in biological treatment processes. These methods may include aerobic and anaerobic treatments, as well as selective microorganisms to degrade specific pharmaceutical compounds [70]. Martins et al. investigated to identify anaerobic microorganisms capable of eliminating pharmaceutical compounds under anaerobic conditions [71]. The target drugs were ciprofloxacin, estradiol, and sulfamethoxazole under nitrite and sulfate-reducing conditions. A total of 80% biodegradation was achieved for ciprofloxacin, while for estradiol it was 84%, and no biodegradation was observed for sulfamethoxazole. The bacteria identified in the decomposition of ciprofloxacin were *Comamonas*, *Archobacter*, *Dysgonomonas*, *Macellibacteroides*, and *Actinomyces*; meanwhile, for the biodegradation of estradiol, *Comamonas* and *Castellaniella* were highlighted. These results indicate the ability of these microorganisms to utilize pharmaceutical compounds as an energy source. Consequently, anaerobic bioremediation emerges as a promising strategy to address the emerging problem of contamination by pharmaceutical compounds in water. It is important to note that the major challenge in microbial fuel cells (MFCs) lies in the operating conditions, such as pH and temperature, which significantly impact bacterial growth.

Electrocoagulation is an electrochemical process that can be effective in removing pharmaceutical contaminants from water by using electrical currents to coagulate and precipitate unwanted compounds, which can then be removed by filtration or sedimentation [72]. The integration of advanced oxidation processes, such as pretreatment and electrocoagulation, emerges as an effective strategy to increase pollutant removal efficiency in wastewater. In particular, the electrocoagulation process combined with electrooxidation, using a boron-doped diamond anode, although expensive, shows a high oxidation potential, leading to the complete mineralization of organic pollutants. In addition, the combination of ozone with electrocoagulation is effective in the degradation of pharmaceutical pollutants, such as ibuprofen. By exploiting the complementary capabilities of electrocoagulation and advanced oxidation, higher removal rates and more effective water purification results can be achieved [73]. However, further research is required to develop stable electrodes.

Ultraviolet (UV) light irradiation can break down pharmaceutical contaminants into less toxic chemicals through photocatalytic oxidation processes, facilitating their removal from water [74]. An anaerobic ammonium oxidation system for nitrite formation by partial nitrification, generated by OH radicals through UV light irradiation in wastewater was studied together to enhance the degradation of the pharmaceutical compounds atenolol, carbamazepine, fluoxetine, and trimethoprim. The degradation of the drugs in wastewater irradiated at a wavelength of 220 nm for 60 min was 69%, 56%, 90%, and 69%, respectively. The combination of these technologies offers a promising strategy to address the presence

of pharmaceutical compounds in wastewater and improve the quality of treated water [75]. In conclusion, treatments, and technologies for the removal of liquid pharmaceutical contaminants are critical to mitigate the negative impacts of these compounds on the environment and protect human health.

3. PW Adsorption under the Circular Economy Paradigm: Biocarbons

Because PW has ecotoxicological effects and is very persistent in the environment, great attention has been paid to developing feasible methods for their sensing and removal. Notwithstanding, the current technologies are not capable of effectively removing this type of micropollutant, and pretreatment is normally required, which makes it costly [14]. Under the paradigm of a circular economy and waste valorization, it has been proposed that the design of absorbents be based on recycled materials. The most known material is biochar, which is thermochemically fabricated under a limited or no oxygen condition environment [76]. To sort out this problem, several approaches have been adopted such as the use of agricultural waste. An extended work was proposed by Puga et al. where a multicomponent pharmaceutical mixture was used to assess the versatility of agroforestry biochar [77]. The biocarbon was fabricated by pyrolysis of grapevine cane, holm three, and eucalyptus residues at 500 °C, which was further sieved to have particle sizes below 75 µm. The obtained sample was used to simultaneously adsorb Venlafaxine, Trazadone, and Fluoxetine into a synthetic batch and real wastewater. The results reveal that the use of biochar is a novel strategy to efficiently remove pharmaceutical mixtures from contaminated effluents. In the same direction, Asati et al. fabricate nitrogen-doped carbonaceous aerogels through jaggery as a carbon source [78]. Ammonium chloride was mixed with the jaggery waste and thermal treated at 150 °C for 2 h. To obtain the final product the sample was further treated at 700 °C under argon atmosphere for 2 h. The adsorption capacity for pharmaceutical waste was assessed by using acetylsalicylic acid and acetaminophen as pollutants. The results show that in only 40 min the developed material delivered an adsorption rate of more than 95% for both pollutants. Magnetic biochar was also tested for the removal of ibuprofen and sulfamethoxazole from water [79]. The carbonaceous material was obtained from the thermal treatment of an orange peel which was modified by incorporating magnetic Fe₃O₄ nanoparticles. The measurements reveal an adsorption capacity of 58.12 and 60.9 mg g⁻¹ for ibuprofen and sulfamethoxazole, respectively. Similarly, biochar was obtained from the thermochemical treatment of rice waste under nitrogen flow with the idea of using it as a pharmaceutically adsorbing material [80]. The electrochemical degradation of Fluoxetine hydrochloride was measured, showing promising results.

It should be mentioned that the ultimate characteristics of fabricated biochars are influenced by the thermochemical synthesis parameters. For instance, in the research work [81] the adsorption capability of cotton gin waste and guayule bagasse was ascribed to the specific surface area and functional groups. Similarly, Akkouche et al. implemented cotton textiles waste, under the paradigm of waste valorization, to fabricate biosorbents via pyrolysis and chemical modification with H₃PO₃ [82]. The cellulosic source was cottoning textile recovered from a second-hand shop which was thermally treated at 600 °C under a nitrogen atmosphere to obtain the carbonaceous material. Then, it was chemically modified under three different concentrations of H₃PO₃. The absorption capability of the obtained material was assessed by testing the absorption of tetracycline and paracetamol.

The adsorption capacity of materials is directly related to their inherent characteristics and can be influenced by several mechanisms. These mechanisms include various processes like pore filling, which involves the diffusion of a substance into the microscopic voids or pores within the material's structure. Additionally, van der Waals forces are another adsorption mechanism where molecules are attracted to one another due to dipoles induced by fluctuations in electron distribution. Furthermore, surface chemistry and physical properties such as surface area, partial charge, and porosity greatly impact the adsorption behavior of materials. Figure 3 presents a diagram that summarizes the most used adsorption mechanisms in material engineering science.

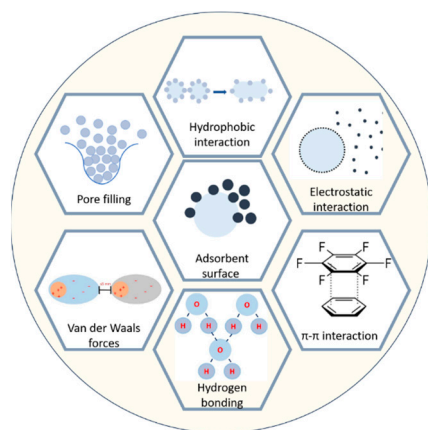


Figure 3. Adsorption mechanisms in material engineering science, based on [83,84].

More recently, Lago et al. reported an outstanding approach where raw pine bark was utilized as a biosorbent for fluoxetine hydrochloride (FLX), carbamazepine, and atrazine [85]. This group proposed surface chemical modification with acid and alkali agents to improve the adsorption capability and, thereafter, hydrothermal carbonization was achieved at 220 °C for 150 min. Surprisingly, the obtained biosorbent showed a fast adsorption rate for fluoxetine hydrochloride in comparison with the other two pollutants. Authors attributed distinct electrostatic interactions between the pharmaceutical molecules and the surface of the treated pine bark, where FLX experiences attraction forces due to differences in surface charges. An interesting work was conducted by Liu et al. where pharmaceutical sludge collected from a pharmaceutical plant in Guangdong, China was mixed with biochar and pyrolyzed at temperatures ranging from 500 °C to 800 °C [86]. During the thermal treatment, the sludge promoted biogas production, making it a promising option for H₂ generation. Additionally, Villamil and colleagues proposed the production of activated carbon through hydrothermal treatment of waste-activated sludge, holding promising potential for adsorbing emerging pollutants such as sulfamethoxazole, antipyrine, and desipramine [87].

4. Photocatalysts Pharmaceutical Wastewater Degradation

The general mechanism of photocatalysis, as seen in Figure 4, is triggered when the semiconductor absorbs photons of energy ($h\nu$) equal to or greater than its band gap, which allows electrons (e^-) in the valence band (VB) to migrate towards the conduction band (CB), thus generating holes (h^+) in the VB. The formation of these e^-/h^+ pairs occurs in the semiconductor, and the photogenerated charges must migrate to the surface to interact with the adsorbed molecules on the photocatalyst. However, before this happens, it is possible for the e^-/h^+ pairs to recombine, dissipating their excess energy as heat, or to become trapped in surface defects [88].

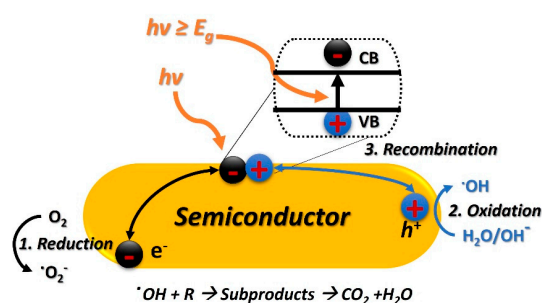


Figure 4. Mechanism of the photocatalytic process in semiconductors.

The majority of photocatalysts employed in photocatalysis are semiconductors that have intricate properties such as their electronic structure and capacity to generate charge carriers upon light absorption. These electronic properties are explained by the band theory which considers semiconducting materials as solids with atoms in a three-dimensional lattice arrangement [89]. A high-quality photocatalyst should possess several key characteristics: non-toxicity, stability, high photoactivity, and cost-effectiveness. Numerous semiconductors exhibit suitable band gap energies for photocatalysis, including TiO_2 [90], ZnO [91], WO_3 [92], Fe_2O_3 [93], and ZnS [94]. These materials have garnered attention due to their ability to efficiently harness light energy for photocatalytic reactions while meeting the criteria. The degradation efficiency of the photocatalytic reaction depends on several factors, such as initial pollutant concentration, type and intensity of irradiation, type and mass of catalyst, pH, temperature, flow rate, and oxygen concentration [95].

4.1. Nanostructured-Based Photocatalysts

Nanostructured photocatalysts are more efficient than bulk photocatalysts due to their surface reactivity. They are composed of globular nanoparticles, nanosheets, nanowires, or nanostructured thin films [96,97]. During the synthesis of these nanostructures, their composition, size, and shape can be manipulated, thus adjusting the photocatalytic properties to obtain higher efficiency and performance in photocatalytic performance. The characteristic nanometer size of the nanostructure facilitates light absorption and charge separation, improving its photocatalytic properties. In addition, its large surface area provides active sites for catalytic reactions and improves their kinetics. Other functionalities include surface coatings by doping with various atoms for increased stability and selectivity [98–100]. For example, Fan et al. studied different morphologies of Bi-doped TiO_2 , titanate nanoribbons, titanate nano-bulks, and titanate nanosheets for acetaminophen degradation, with the nanoribbon morphology being the one that showed the best photocatalytic performance with a degradation under visible light of 88% at an initial concentration of acetaminophen (ACM) 0.7 mg L^{-1} for 180 min [101]. Thirteen intermediates formed during the degradation process were identified in this study. This same research group worked with Bi-doped TiO_2 nanobulks for the degradation of naproxen with a concentration of 0.25 mg L^{-1} obtaining 99% degradation in 108 min [102].

4.2. Metal-Based Photocatalysts

Metal-based photocatalysts, which incorporate metallic elements in their structure or surface, have enhanced photocatalytic activity and stability due to the special properties of metals and their interactions with light and reactive species [103]. These photocatalysts can exist in different forms, such as metal nanoparticles, metal oxides, metal sulfides, metal complexes, and metal-doped semiconductors [104,105]. Different synthesis methods can be used to create these materials, such as sol-gel, hydrothermal synthesis, chemical vapour deposition, and impregnation techniques [106,107]. The photocatalyst TiO_2 , co-doped with Pt and Bi under visible light irradiation, has been studied for the degradation of amoxicillin (AMX) at a concentration of 10 mg L^{-1} , obtaining an efficient degradation of 87% in a reaction time of 120 min, according to results reported by Salimi et al. [108]. Al-Musawi et al. achieved 100% degradation of AMX at a concentration of 25 mg L^{-1} under UV-light irradiation for 60 min, with a $\text{Fe}_2\text{O}_3/\text{B}/\text{TiO}_2$ photocatalyst which showed an 8% reduction in efficiency after six cycles of reuse [109]. Meanwhile, $\text{Fe}_3\text{O}_4@\text{void}/\text{CuO}/\text{ZnO}$ has also been studied in the presence of visible light for the degradation of AMX. In this case, a degradation efficiency of 97.6% was obtained with an initial AMX concentration of 30 mg L^{-1} , showing high stability in the degradation efficiency after five cycles of reuse [110]. Similarly, FeCe/SiO_2 S-scheme photocatalyst was explored to degrade levofloxacin using a response surface methodology where 30 trial experiments were conducted to construct the model considering four factors: levofloxacin concentration, photocatalyst dosage, time, and pH [111]. The experimental results demonstrate that 30% FeCe/SiO_2 demonstrated the highest mineralization efficiency of 69%. This compound was also tested

for tetracycline degradation and the result revealed an efficiency of 94%, which is superior to that reported by Chatterjee [112]. On the other hand, Mirzaei et al. [113] report the degradation of sulfamethoxazole (SMX) at a concentration of 30 mg L^{-1} by a magnetic $\text{ZnO@g-C}_3\text{N}_4$ composite obtaining a degradation efficiency of 99% for 60 min, in addition to the removal of 45% of total organic carbon (TOC). Meanwhile, Lien et al. [114] report SMX degradation on $\text{CaCu}_3\text{Ti}_4\text{O}_7$ perovskite with $\sim 99\%$ degradation efficiency under UV-light irradiation over a 90 min period. Interesting work was reported by Machín where $\text{Co}_3\text{O}_4\text{-gC}_3\text{N}_4$ -coated ZnO Nanoparticles were used as photocatalysts for ciprofloxacin degradation [115]. The process involved a mix of the composite (1.1 g L^{-1}) with 10 mol of the drug where the pH was regulated by adding NaOH or HCl. Next, H_2O_2 was added, and the solution was aerated before being exposed to simulated sunlight. Figure 5 depicts the photodegradation results for ciprofloxacin over a 60 min period. Initially, a high concentration of $\text{Co}_3\text{O}_4\text{-gC}_3\text{N}_4$ significantly enhanced the degradation process, achieving over 80% degradation. However, at a 10% composite concentration, the photodegradation rate decreased, likely due to a saturation effect.

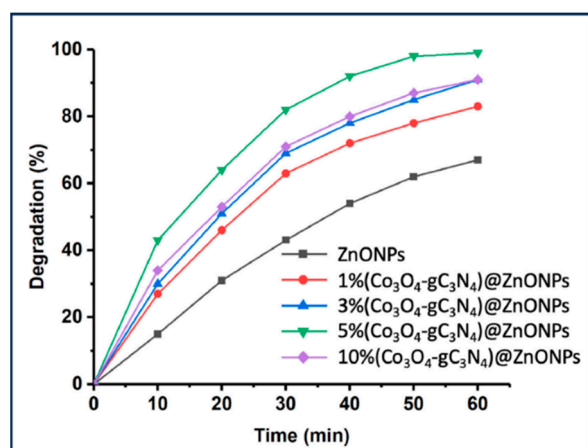


Figure 5. The observed rate of CFX (10 mM) photodegradation and catalysts loading of 1.1 g L^{-1} across various composites at different concentrations. The solution was then exposed to a solar simulator, equipped with dual white light bulbs (100 watts and ca. 6300 lux. Machín et al. 2024[®], Reproduce under Creative Commons Attribution License (CC BY) <https://creativecommons.org/licenses/by/4.0> (accessed on 7 May 2024) [115].

Tetracyclines (TCs) are broad-spectrum antibiotics that are recommended for the treatment of bacterial infections [116]. In addition, these pharmaceuticals are also implemented in the agricultural sector as soil fertilizers, raising significant environmental and health concerns [117]. In this context, photocatalysts might be a viable strategy not only for treating water contaminated with antibiotics but also for producing valuable products such as H_2 . Tungsten oxide embedded in $\text{BiOCl/g-C}_3\text{N}_4$ was utilized as a photocatalyst to degrade TC antibiotics in an aqueous solution [112]. The simulated light was created by a xenon lamp of 150 W with a 520 nm wavelength while the drug concentration was determined by UV-Vis. Surprisingly, the developed material shows degradation activity both in the presence and absence of simulated sunlight. After just 20 min of light exposure, it achieved an 85% degradation of TC. In addition, the same composite was tested for hydrogen evolution, with methanol added as a sacrificial molecule in this case. In another work, Qi et al. reported the fabrication of $\alpha\text{-NiS/g-C}_3\text{N}_4$ nanocomposite by two consecutive thermochemical processes which were then used as photocatalysts for TC degradation and also for hydrogen production [118]. The experimental H_2 production was achieved by mixing 10 mg of the nanocomposite with an aqueous solution containing triethanolamine and eosin yellow acid dye as a sacrificial agent. This mixture was then transferred to a sealed reactor, exposed to a 300 W Xe lamp, and purged with N_2 . Figure 6

illustrates the result obtained by these researchers, highlighting the outstanding efficiency of H_2 generation achieved with 15% α -NiS/g- C_3N_4 .

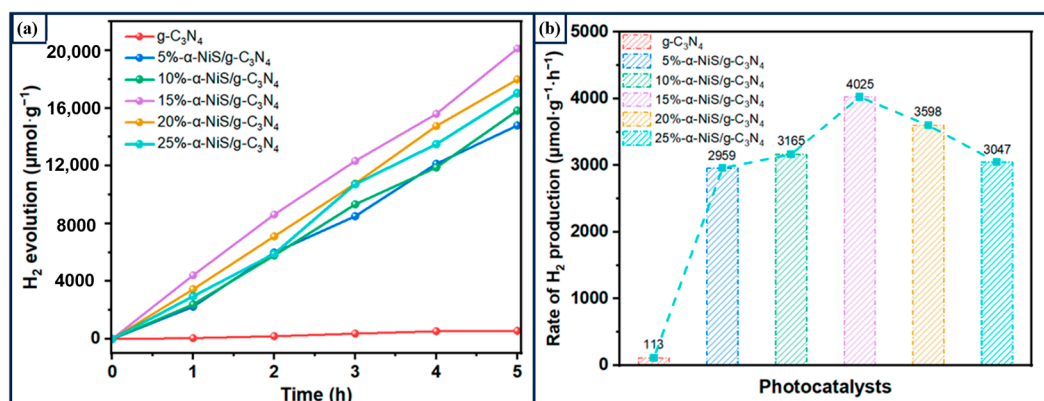
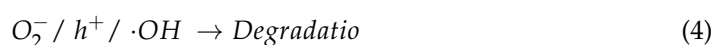
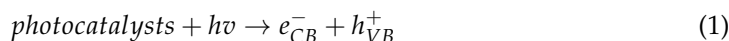


Figure 6. Hydrogen production activity of the synthesized material. (a) Experimental curve for each nanocomposite; (b) rate of H_2 production for each nanocomposite. Qi Huajin 2023[®], Reproduce under Creative Commons Attribution License (CC BY) <https://creativecommons.org/licenses/by/4.0> (accessed on 15 May 2024) [118].

An interesting study was reported by Lin and colleagues, who fabricated a composite photocatalyst, $\text{Ag}_3\text{PO}_4@\text{W}_2\text{N}_3\text{-NV}$, for the removal of β -lactam antibiotics [119]. The same research group also synthesized the $\text{Ag}_3\text{PO}_4@\text{NC}$ photocatalyst and tested its effectiveness in degrading three emerging pollutants: norfloxacin, diclofenac, and phenol [120]. The results demonstrate that this compound is capable of achieving rapid degradation under visible light. For instance, the photocatalyst exhibited 100% efficiency in degrading norfloxacin in just 5 min. Similarly, another system with Ag_3PO_4 is the $\text{Ag}_3\text{PO}_4@\gamma\text{-G}$ heterojunction, which achieves a photocatalytic degradation rate of norfloxacin (NFL), 2-hydroxynaphthalene (2-HNP), and phenol, reaching 100% in only 8.5 and 16 min with visible light irradiation, with rates greater than 15.3, 9.6 and 19.7 times higher than when using Ag_3PO_4 [121].

The reports mentioned above highlight the potential of photocatalysts for use in both degradation processes and hydrogen production. A significant breakthrough in the photocatalyst field was reported by Wang et al., who demonstrated the simultaneous degradation of antibiotics alongside H_2 evolution (using the antibiotic as an electron acceptor) [122]. A $\text{Bi}_3\text{TaO}_7/\text{ZnIn}_2\text{S}_4$ composite was tested as a photocatalyst under visible light with a 420 nm wavelength. The detailed study shows that h^+ and OH^- were the active species for the TC degradation process. The system with H_2PtCl_6 shows accelerated H_2 evolution at the initial increase in tetracycline concentration, but a change in the reaction rate occurs when high drug doses are used, particularly at concentrations exceeding 40 mg L^{-1} . The accepted mechanism for simultaneous pollutant mineralization along with H_2 evolution is explained by the following equations [23].



In the same direction, Shang and their coworkers employed BaTiO₃/Ag₂S nanofibers to achieve HER coupled with TC photodegradation in a sealed three-necked glass cell irradiated with a 350 W xenon lamp [123]. The experimental procedure consisted of mixing 10 mg of the composite in 100 mL of TC aqueous solution (40 ppm), followed by deaeration using N₂. Samples were collected hourly and analyzed using chromatography. After the test, the degraded solution was measured using ultraviolet spectrometry. The composite delivered a degradation efficiency of 84.4% in 90 min with acceptable hydrogen evolution activity, 597 mol h⁻¹g⁻¹. The authors concluded that the improved performance was ascribed to: (a) the broad light absorption spectrum, and (b) the close heterojunction interface between BaTiO₃ and Ag₂S, facilitating the electron migration path and then enhancing the separation and transfer of charge carriers. A NiFe-CO₃²⁻-LDH/Sn⁴⁺-β-Bi₂O₃ microsphere was also tested for TC degradation [124], exhibiting promising results due to the large hydrogen activity of 1851.01 μmol h⁻¹g⁻¹, which was larger than reported by Shang [123]. This improvement was mainly attributed to the beneficial effect of Sn⁴⁺ incorporation, which led to an enhanced UV-Vis response, extended carrier lifetime, and reduced surface resistance.

4.3. Metal-Organic Photocatalysts

Metal-organic framework (MOF)-based photocatalysts are a type of advanced material that combines the properties of metal-organic frameworks and photocatalysts. MOFs are highly ordered crystalline structures composed of metal ions or clusters coordinated with organic ligands, resulting in a large surface area and adjustable pore size. By incorporating photocatalytically active components, such as semiconducting nanoparticles or organic chromophores, into the MOF structure, synergistic effects can be achieved between the MOF structure and photocatalytic properties [125,126]. The porous nature of MOFs provides abundant active sites for catalytic reactions, while the ordered framework structure enables efficient light absorption and charge separation. MOF-based photocatalysts have shown promise in various environmental and energy-related applications, including pollutant degradation, water splitting for hydrogen generation, carbon dioxide reduction, and organic synthesis [127]. The tunable chemical and structural properties of MOFs allow for customization and optimization of their performance for a wide range of photocatalytic applications [128]. D. Ali et al. developed a ternary photocatalytic heterojunction composed of carbon-doped TiO₂ (C-TiO₂) decorated with Zn-Zeolitic imidazole framework (ZIF-8) and AgCl for application in the degradation of the antibiotic levofloxacin (LVFX); the recombination and synergy of the components resulted in 98% LVFX degradation for 60 min under solar illumination [129]. Meanwhile, Wang et al., studied the degradation of the antibiotic TC at a concentration of 20 mg L⁻¹ using a NiIn₂/UiO-66 heterojunction photocatalyst obtaining 90% degradation under visible light irradiation for 1 h [130]. On the other hand, Solis et al. prepared NH₂-MIL-125(Ti) which showed the best performance in the removal of diclofenac with 99% over 3 h under solar simulator irradiation [131]. Concluding that the design of heterojunctions provides a cost-effective strategy for the remediation of pharmaceutical-compound-contaminated water. On the other hand, covalent organic frameworks have also been proposed for the fabrication of active photocatalysts. For example, COF@H₃PO₄ was synthesized and tested as a photocatalyst for the H₂O₂ generation without a sacrificial agent, achieving an efficiency of 0.69% [132].

4.4. Non-Metal-Based Photocatalysts

Non-metallic photocatalysts are materials that can activate photocatalytic reactions when exposed to light, even though they do not contain metallic elements in their composition. These materials are usually composed of non-metallic elements such as carbon, nitrogen, sulfur, phosphorus, or non-metallic compounds. Some examples of non-metallic photocatalysts are graphene, carbon nanotubes, and carbon nitride (g-C₃N₄), which stand out for their electronic properties, high surface area, and stability, making them suitable for applications in hydrogen production, pollutant degradation, and carbon dioxide reduc-

tion [133,134]. Non-metallic organic compounds, such as porphyrins, conjugated polymers, and organic dyes, are also used as photocatalysts, since they can absorb visible light and generate photoinduced reactions, such as hydrogen evolution, pollutant degradation, and organic synthesis [135]. Liu et al. [136] studied the ability of single/few-layered carbon nitride nanosheets with carbon vacancies (Cv-CNNs) against the antibiotic sulfadiazine (SDZ), achieving 100% degradation efficiency of SDZ at a concentration of (5 mg L^{-1}) for 20 min under visible light irradiation. The study provides a metal-free photocatalyst for application in the removal of antibiotics from wastewater. On the other hand, Balarak et al. used graphene oxide with titanium oxide nanoparticles (GO/TiO₂) for the degradation of AMX in an aqueous solution of 50 mg L^{-1} , reporting 99% degradation during 60 min under UV-light irradiation [137]. The degradation process showed stability and recyclability of GO/TiO₂, as well as complete mineralization of the contaminant and its products. The advantages of using photocatalysts for the degradation of pollutants in wastewater are wide applicability and effectiveness for the wide range of organic pollutants present in pharmaceutical compounds. On the other hand, in many cases, the use of additional additives is not required, which reduces costs and possible hazardous by-products. In this way, it has a low environmental impact as the process leads to the formation of non-toxic by-products, minimizing the negative environmental impact of wastewater treatment [138]. However, there are challenges to consider: the efficiency of the photocatalytic degradation can be influenced by factors such as the type of elements in the photocatalyst, the wavelength of light, and the concentration of contaminants. In addition, it should be considered that, under specific conditions, the photocatalytic process can generate by-products that need to be carefully monitored for their potential negative effect on the environment and, ultimately, their costs, as implementing a photocatalyst on a larger scale can be a limiting factor [139]. Activated sewage sludge has demonstrated good performance as a substrate for TiO₂ photocatalysts in the degradation of TC [140]. The high degradation rate, which was four times greater than that of untreated TiO₂, was attributed to the presence of metal and non-metal elements in the sludge. These elements doped the TiO₂ structure, reducing its bandgap. Another interesting approach was reported by Guz et al., who integrated TiO₂ and activated sludge for the treatment of industrial effluent red water, conformed by sulfonates and other aromatic compounds [141]. Interestingly, this compound shows a reduction of 94% for nitro aromatics and 72% reduction in total phenols.

5. Microbial Fuel Cell for PWW Treatment

Microbial fuel cells (MFCs) are bioelectrochemical devices that have been developed as an alternative for wastewater treatment and electricity generation. For these purposes, MFCs take advantage of the metabolism of some microorganisms (electrochemically active microorganisms) capable of transforming the chemical energy present in the organic compounds into electrical energy; this is done by transferring the electrons produced during microbial metabolism to an electrode that functions as an external electron acceptor (anode), thereby reducing the content of organic compounds that pollute wastewater [142–144].

In general, a typical dual-chamber MFC is composed of an anode containing a bioanode (anode + microorganisms), a cathode compartment, a proton exchange membrane separating the two chambers, and an external electrical circuit [145]. Figure 7 displays the configuration of the dual MFC where the chemical energy stored in the PW is directly converted into electrical energy.

On the other hand, dependence on drugs and therapeutic products has increased considerably due to the prevalence of numerous diseases. This has, in turn, caused a massive rise in the production of drugs and release of effluents loaded with pharmaceutical contaminants, so that polluted water, has become a serious environmental problem due to its potential health effects, its nature, persistence, and resistance to degradation [26]. For example, antibiotics are compounds that do not change their structure and composition when disposed of in sewage systems, which can cause pathogenic microorganisms to generate resistance to antibiotics, thus modifying the structure of microbial communities in

ecosystems [146]. For this reason, new technologies such as MFCs have been developed to treat contaminated water and produce electricity sustainably, and therefore research has focused on the evaluation of the efficacy and performance of MFCs for the degradation of pharmacological grade organic compounds. In this sense, different substrates (PWW), diverse electrode materials for both the anode and the cathode, and different cultures of individual microorganisms in the form of consortia with exoelectrogen characteristics and capable of generating biofilms on the anode, have been studied. PWW represents a great challenge to its treatment since it is characterized by a complex composition, high toxicity, and many organic constituents, which represent a high chemical oxygen demand (COD). Ismail and Habeeb [147] evaluated the efficiency of a double-chamber MFC for the treatment of PWW and power generation, using a granular activated carbon anode, obtaining a COD removal efficiency of 83% and a power density of 204.9 mW m^{-2} . On the other hand, Alonso-Lemus et al. [142] reported a COD removal of 45% from PWW and a maximum power density of 96.3 mW m^{-2} using a double-chamber MFC and an anode composed of a biocarbon synthesized from leather waste supported on carbon felt, and a biofilm formed by the bacterium *Bacillus subtilis*.

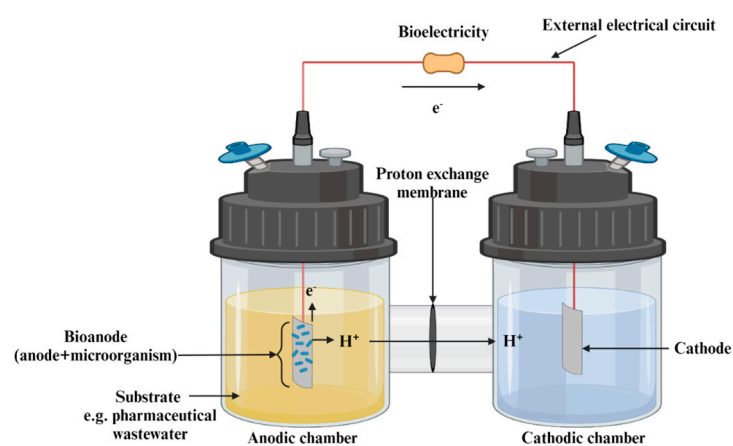


Figure 7. Typical representation of a dual-chamber microbial fuel cell.

On the other side, the application of MFC has been described for the elimination of a specific drug present in PWW. For example, the use of MFC has been reported for the elimination of sulfonamide-type antibiotics, where a synthetic substrate consisting of porcine wastewater conditioned with SMX, sulfadiazine, and/or sulfamethazine was used. In this case, a double-chamber MFC was employed, using in the anode chamber an anode composed of a biocarbon synthesized from a grapefruit peel supported on graphite felt, a microbial inoculum based on sludge from an anaerobic digester, and, in the cathode chamber, a carbon fiber brush which was used as the cathode; degradation efficiencies of antibiotics were obtained, ranging from 82.44 to 88.15%, 53.40 to 77.53% and 61.12 to 80.68% of SMX, sulfadiazine, and sulfamethazine, respectively, and, in addition, they obtained a decrease in the COD of more than 98% [148]. To enhance the degradation efficiency of sulfamethoxazole, Xu et al. employed Microbial Fuel Cell-constructed wetlands that contain Firmicutes, Proteobacteria, and Bacteroidetes. Their approach achieved an impressive 93.6% efficiency in degrading the molecule [149]. Similarly, Al-Ansari et al. have reported on the degradation of sulfadiazine present in wastewater, and in their study they used a double-chamber MFC, a cylindrical anode of graphite felt, a carbon fiber brush as cathode and the bacterium *B. subtilis* as electrochemically active microorganism; in this case, they reported a 95.3% removal of sulfadiazine because *B. subtilis* can degrade some antibiotics. They obtained a COD degradation efficiency of 91.9%, so they conclude that it is of utmost importance to make a good selection of the microorganisms used in the MFC to improve the efficiency of the degradation of organic pollutants, such as pharmaceuticals [150].

Wang et al. reported on the successful degradation of TC using a bio-cathode in an MFC [151]. A cylindrical two-chamber MFC reactor was constructed, utilizing carbon felt as the material for both the anode and cathode. The anode chamber was filled with a mixture that included glucose, phosphate-buffered saline, vitamins, and other essential compounds, while the cathode chamber contained sodium bicarbonate, sodium acetate, phosphate-buffered saline, and additional components. The inoculated sludge was obtained from a treatment plant in Guangzhou, Guangdong province. Figure 8 presents the degradation rates for each tested process. It might be noted that MFC bio-cathode degradation has the highest rate, 88.86%, which confirms the effectiveness of this technology in degrading drug waste. The authors suggested that the observed enhancement could be attributed to the small electrical stimulation of MFC cells, which favors the growth and metabolism of microorganisms. This, in turn, leads to a significant improvement in degradation efficiency.

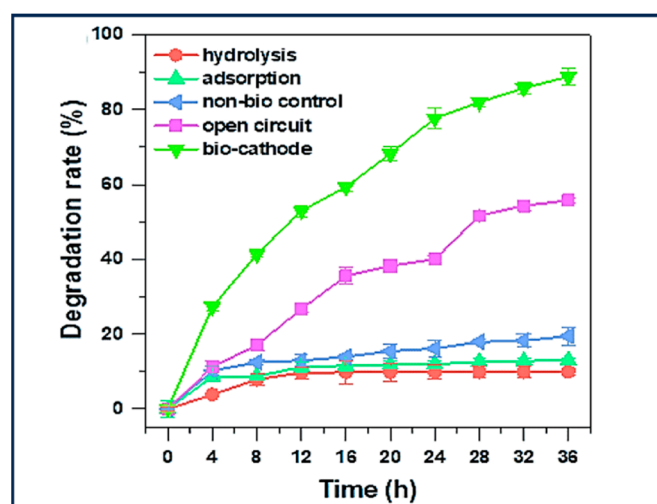


Figure 8. Degradation process on MFC bio-cathode. Wang et al. 2022[®], Reproduce under Creative Commons Attribution-NonCommercial 3.0 Under supported Licence [151].

Activated sludge was also integrated with a microalgae fixed film to construct a new MFC system, achieving an average removal of 82% for chemical oxygen demand and 90.3% for $\text{NH}_4^+\text{-N}$. [152]. Recently, Gu et al. explored the implementation of residual sludge in MFC for the treatment of simulated chromium wastewater [153]. The residual sludge, collected from a secondary sedimentation tank at a sewage treatment plant in Nanjing, was anaerobically fermented for seven days and then used as inoculum for the MFC. The system achieved a maximum power density was $100.52 \text{ mW cm}^{-2}$ with a peak output voltage of 624 mV. Another type of drug of interest, present in wastewater, is non-steroidal anti-inflammatory drugs (NSAIDs), which present incomplete degradation in treatment systems as well as having the capacity to promote physiological problems in animals and humans [154]. Morovati et al. reported a diclofenac degradation of up to 56% and a COD removal of 94.67%, using an MFC with a MnCo_2O_4 anode supported on carbon felt and adding sludge from an anaerobic digester as bacterial inoculum [155]. Qiu et al. report a diclofenac degradation performance of 75.59% by a single-chamber MFC using a carbon felt anode loaded with a Ru/Fe alloy and sludge from an anaerobic reactor as microbial inoculum where bacteria of the genus *Geobacter*, *Clostridium*, *Sedimentibacter*, *Pseudomonas* and *Desulfovibrionaceae* were found to be able to degrade diclofenac [156]. Therefore, enriching substrates with functional microorganisms can contribute significantly to the degradation of drugs present in wastewater, facilitating their treatment; however, the physicochemical characteristics (hydrophobicity, solubility, composition) and concentration of the drugs in the wastewater must be considered, since they can be factors that limit or favor their elimination. The main drawback of MFC scale-up is the cell voltage reversal that occurs due to the imbalance of the stacking cells [157]. However, cost-effectiveness

during scale-up should be considered, with estimates ranging from \$735 per cubic meter to \$36,000 per cubic meter for large stacks. Therefore, it is suggested to continue research on the manufacture of materials for the elaboration of low-cost electrodes that promote microbial affinity. In addition, gaining a detailed understanding of the metabolic processes of microorganisms used in MFCs is essential for selecting those with electrochemical activity. This knowledge also aids in refining or adapting the architectural design of the cells and assessing their long-term performance.

Table 2 provides a detailed comparison of pharmaceutical degradation efficiency across various MFCs. Additionally, it includes information on the specific types of drugs being targeted and the corresponding MFC configurations used in the study.

Table 2. Degradation of different drugs using MFC.

Type of Drug	MFC Configuration	Treatment Efficiency	Reference
Oxytetracycline	Dual-chamber MFC with carbon felt anode and cathode	99% elimination	[158]
Cefazolin sodium	Single-chamber MFC with activated carbon air cathode and carbon felt anode.	>70% elimination	[159]
Ibuprofen	Dual-chamber MFC with PANI@CNT-coated stainless steel mesh anode and CuInS photocathode.	75.94% removal	[145]
Sulfamethoxazole	Dual-chamber MFC with carbon felt anode and carbon cloth cathode loaded with 20% Pt.	83.3%	[160]
Carbamazepine	Single-chamber MFC, with carbon cloth cathode in air atmosphere, the anodes consisted of fish bone biocarbon synthesized in air atmosphere (BCA) and another in nitrogen atmosphere (BCN) supported on a stainless-steel mesh.	BCA: 77.88% BCN: 79.58%	[161]
Wastewater from non-steroidal anti-inflammatory drug-producing industry	Dual-chamber MFC; stainless steel anode and Pd-coated Ti or Pd/Ir-coated Ti cathode.	93% COD removal using the Pt-coated Ti cathode and 91% COD removal using the Pd/Ir-coated Ti cathode	[162]
Synthetic water with diclofenac sodium	Dual-chamber MFC; stainless steel anode and Pd-coated Ti or Pd/Ir-coated Ti cathode.	78% COD removal using the Pt-coated Ti cathode and 71% COD removal using the Pd/Ir-coated Ti cathode.	[162]
Sulfamethoxazole	Dual-chamber MFC, graphite fiber anode and stainless-steel cathode with activated carbon.	98.4%	[163]
Sulfamethoxazole	Single-chamber, carbon fiber mesh as anode and carbon cloth coated with Pt/C as cathode.	95.7% at 40 mg L ⁻¹ drug concentration	[164]
Tetracycline	Single-chamber air cathode, anode was a carbon fiber washed with acetone, cathode carbon cloth coated with Pt/C.	84.9% containing 10 mg L ⁻¹⁰ drug	[165]
Tetracycline hydrochloride	Dual-chamber microbial fuel cell configuration, carbon felt as anode and cathode.	>94%	[166]
Sulfadiazine and sulfamethoxazole	Single-chambered cylindrical soil MFCs, granular activated carbon and carbon felt as anode and cathode.	>57%	[167]
Sulfamethoxazole	Microbial fuel cell-constructed wetlands, Coke layer as anode and cathode.	93.6%	[149]
Diclofenac sodium	Two-chamber MFC, anode and cathode of graphite felt.	Up to 30.73% removal in only two weeks of operation	[168]

6. Challenges, Anticipating Actions and Future Perspective of PWM

As stated before, PW is a serious threat to ecosystems and human health, thus, improper disposal is one of the biggest challenges that needs to be addressed to reduce these kinds of pollutants in our environment. The intervention of Governmental institutions is essential to the correct management and re-use of PW where the development of well-established economic incentives and punitive international regulations will be mandatory. Historically, economic incentives have been demonstrated to encourage responsible behavior among pharmaceutical companies, healthcare facilities, and residents [169]. Tax breaks or subsidies for big industries might be the strategy to motivate these entities to invest in eco-friendly waste management practices. In this sense, Derhab et al. conducted a critical review where they analyzed tax relief, subsidies, and even soft loans; the latter strategies not only might reduce waste but may also minimize production costs for manufacturers [170].

On the other side, while economic incentives provide positive reinforcement, punitive regulations are indispensable in avoiding negligent conduct. In this direction, Li et al. reported a complete study of social norms and punitive measurements for enterprises [171]. They studied several policy scenarios as a measurement of changing behavior in the absence of economic benefits and the result was that this approach promoted the diffusion of cleaner production technologies. They prefer to invest in new technologies that lose money as punishment.

Regarding the proposed strategies, there is a lot to do in the area of PW management. Collaboration between governments, regulatory entities, industry stakeholders, and environmental organizations will be essential for designing and implementing effective policies that guarantee the correct disposal and re-use of emerging pollutants. This work studied in detail the most sophisticated methods for PW treatment. Based on the findings, some recommendations and future considerations are proposed. In the near future, there is a necessity to understand the stability of materials with photodegradation capabilities. Simple yet effective materials, such as oxygen-doped carbon nitride, should be considered with a focus on favoring the hydroxyl peroxide pathway, as this can reduce the energy barrier for H_2 generation [172]. Additionally, researchers should explore alternative, non-conventional technologies capable of producing hydrogen as a valuable byproduct from wastewater valorization. One promising technology is aqueous phase reforming, a process that can easily convert biomass dissolved in water directly into hydrogen and CO_2 [173]. The mentioned process is advantageous considering that the process occurs without water vaporization, making it both energy-efficient and potentially more sustainable. Although these technologies demonstrate high efficiency in degrading pharmaceutical pollutants more sustainably, their cost-effectiveness remains closely tied to ongoing research and development. Conducting a detailed cost analysis within specific contexts is required to fully understand their feasibility.

Novel approaches such as machine learning are very promising, offering insights into the mechanisms of photodegradation processes. Through the integration of machine learning algorithms, researchers can identify the critical parameters that influence material stability, predict degradation pathways, and devise targets. Furthermore, the adoption of machine learning enables the development of predictive models that can anticipate material behavior under harsh environmental conditions, thus facilitating the design of more durable materials. Cost analysis is another fundamental approach that needs to be considered during the researching of material stability. As is well-known, society has increased generation of emerging pollutants, so researchers must create cost-effective strategies for mitigating photodegradation and enhancing material stability.

Finally, it is important to highlight the potential for PW to generate valuable by-products such as hydrogen, oxygen, and even electrical energy. This approach could incentivize pharmaceutical industries to adopt a circular economy framework, thereby enhancing their revenue streams.

7. Conclusions

It is evident that PW is a real challenge for society since large industrialization contributes to the generation of emerging pollutants. Additionally, the improper disposal of commonly used medications is a critical issue, as it contaminates the ecosystem. In this regard, this work delivers a comprehensive overview of the current status and future of PW valorization. It visualized novel strategies to face the imminent problem of pharmaceutical waste. For example, biochar has the potential to significantly contribute to the removal of emerging pollutants, while promoting a circular economy. Photocatalysts are promising due to their ability to degrade active compounds in PW water. This technology is easy to adopt and uses low-cost and available materials called semiconductors. Herein, it should be highlighted that, during photocatalytic degradation, it is possible to obtain valuable by-products such as H_2 and O_2 , which can be used as fuel for other processes. On the other side, microbial fuel cells seem to be a feasible pathway for addressing pharmaceutical waste while simultaneously producing electrical energy; however, this device requires more research to find stable and active electrocatalysts. The main obstacle to scaling up these technologies is their high cost, which is significantly limiting their broader adoption and widespread implementation. Nonetheless, it is expected that biochar and photocatalyst technologies will soon see widespread adoption in practical applications, driven by the growing need for effective water treatment solutions.

Last but not least, machine learning and artificial intelligence will help in the development and optimization of advanced materials by using the generated data approach. The main challenges for PW management are the creation and dissemination of international laws and regulations that help in the adoption of sustainable technologies for emerging pollutant treatment. In this sense, a lot of work is required to create a feasible pathway toward both efficient electrical energy production and strategies to degrade pollutants in freshwater systems.

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